Measurement and Analysis of Pulse-oxi and ECG signals

Course: University of Twente Module 5 TN: Signals (202000691)
Course: University of Twente Module 5 AT: Signals (202000627)

1 Introduction

This assignment is on the measurement of some real-life signals and an appropriate treatment of these signals. You will be measuring two signals from your body using two techniques:

- pulse oximeter photoplethysmograph (pulse-oxi or pleth in short) and
- electrocardiogram (ECG).

The advantage of the use of pulse-oxi and ECG signals is that these signals have recognizable content, however they also contain a lot of artefacts that might influence the interpretation. You can find a short description of the pulse-oxi and ECG techniques in Appendix A. You will be applying the signal analysis (Fourier transform, correlation) and processing (filtering) techniques that you learned during the module with the following intended learning objectives:

- Applying Fourier transform to real-life signals
- Interpreting real life signals in the frequency domain
- Basic quantification of signal-to-noise ratio (SNR) of signals
- Designing FIR and/or IIR filters against a criteria and justifying choices
- Applying auto- and cross-correlation techniques
- Interpretation of correlation between two signals and two techniques (pulse-oxi and ECG)
- Learning the software packages and commands in MATLAB and PYTHON for processing signals

This assignment consists of three phases.

- **Phase 1** - Recording of the pulse-oxi and ECG signals in the lab: These signals will be measured on live human beings: you and a partner. You are expected to record your signals to your computer, which you will be analysing and treating in the next phases of the assignment.
- **Phase 2** - Preliminary analysis of the raw data: You are expected to deliver a short report on your analysis. This **intermediate report** will form 35% of the ECG assignment grade.
- **Phase 3** - Final analysis of the signals: You are expected to deliver a more elaborate final report on your signal analysis and treatment. This **final report** will form 65% of the ECG assignment grade.

You can find the submission deadlines on Canvas website.

2 Assignment

2.1 Phase 1 - Pulse-oxi and ECG data recording

For recording your pulse-oxi and ECG signals, you have to sign up in pairs through the website https://practica.tnw.utwente.nl//tn. You will be using a pulse-oxi meter and a ECG recording setup in the lab. You can find the details of the equipment, software, and measurement procedure in the Appendices B and C.

Using the instructions in the appendices, record your (or your partner's) pulse-oxi and ECG signals for 40 seconds. You should record the following 4 signals:

- Oxygen saturation signal from your finger clip (pleth) – *sig-P*
- Electric signal from the right wrist: – *sig-R*
- Electric signal from the left wrist: – *sig-L*
- Electric signal from the right ankle: – *sig-F*

To make sure that your recording is good and retrievable afterwards, you are expected to show your plotted raw data using MATLAB, PYTHON or EXCEL to one of the TAs or profs in the lab. Note that the electric signal from the right ankle is used as reference for the other electric signals, and does not appear in the data you retrieved from the data file.

2.2 Phase 2 - Preliminary analysis - Intermediate report (35 points)

This phase is about performing a preliminary analysis of the data you recorded and submitting it in the form of a short report. The report should consist of 4 sections following the 4 sections of the assignment.

2.2.1 Raw data

a. Present the measured four signals in four sub-figures. Limit the presented range to 10 seconds.

b. Comment on the prominent features you observe in 2-3 sentences.

2.2.2 Fourier transform of the raw data

a. Perform a Fourier transform on the four signals. Present the amplitude of the frequency spectrum in four sub-figures. Limit the frequency axis to 0–80 Hz range.

b. Comment on the prominent peaks in your spectra in 2-3 sentences.

c. Comment on the comparison of four spectra.

2.2.3 Auto-correlation

a. Calculate the auto-correlations of four signals. You are advised to limit your auto-correlation calculations to a 5 seconds long sub-part of the signal that is around 0 and showing least drift. Plot the results in four sub-figures. Comment on your observations for each sub-figure.

b. Compare the signals and comment on your observations.

2.2.4 Cross-correlation

a. Calculate the cross-correlation between *sig–P* and the cleanest signal out of the three ECG recordings. You can again limit your correlation calculations to a 5 seconds long sub-part of the signal that is around 0 and showing least drift. Plot the result in a figure. Comment on your observations in 2-3 sentences.

b. Compare your cross-correlation observations to your auto-correlation observations.

c. Conclude with a combined interpretation of the Fourier transform, auto-correlation and cross-correlation results.

2.3 Phase 3 - Final analysis - Final report

This phase is about performing a more advanced analysis and processing of the data you recorded. You will be submitting your analysis in the form of a final report. The report should consist of 5 sections following the 5 sections of the assignment.

2.3.1 SNR of raw data

For an explanation of SNR, you can refer to the Canvas page for Learning Task on SNR estimation.

a. Calculate the SNR for all four signals and tabulate them. Show using a figure how you calculated SNR. Comment on the important points to pay attention in calculating SNR.

2.3.2 Design and application of high-pass filter

a. Design a high-pass filter to straighten the base line of the raw ECG signal with the highest SNR. Clearly motivate the choices you made in the design of the filter.

b. Present the filter characteristics in an annotated graph in the frequency domain.

c. Apply the filter on your selected ECG signal.

d. Present the filtered signal in the time-domain as well as in the frequency domain. Comment on the changes you observe with respect to unfiltered signal.

e. Calculate the SNR of the filtered signal and append them to your previous SNR table. How does the SNR change after filtering? Comment on the influence of filter parameters to the changes in the SNR.

2.3.3 Design and application of low-pass filter

a. Design a low-pass filter to suppress the noise in the high-pass filtered signal. Carefully choose your parameters in such a way as not to modify the shape of the relevant signals too much. Clearly motivate the choices you made in the design of the filter.

b. Present the filter characteristics in an annotated graph in the frequency domain.

c. Apply the filter on your high-pass filtered signal.

d. Present the filtered signal in the time-domain as well as in the frequency domain. Comment on the changes you observe with respect to unfiltered and high-pass filtered signals.

e. Why are you allowed to apply the high-pass filter and the low-pass filter in two successive steps? Would a single band-pass filter give you advantage in performance?

f. Calculate the SNR of the low-pass filtered signal and append to your SNR table. Compare your new SNR to the unfiltered and high-pass filtered SNR values.

2.3.4 Auto-correlation

a. Calculate and plot the auto-correlations of the filtered signal. Comment on your observations in 2-3 sentences.

b. Compare your new correlation result with the unfiltered case in Phase 2.

2.3.5 Cross-correlation

a. Calculate and plot the cross-correlation between $sig–P$ and the filtered signal.

b. Compare your new correlation result with the unfiltered case in Phase 2.

c. Compare your cross-correlation observations to your auto-correlation observations. What are your new observations beyond Phase 2?

d. Conclude with a combined interpretation of the Fourier transform, auto-correlation and cross-correlation results.

Appendix A ECG technique

The idea of the ECG-measurements is that you measure the electric signals that are generated by your heart. These signals can be measured by picking up the electric potential across your heart with electrodes on your skin. We use the so called *Einthoven*-method, named after Willem Einthoven who invented the method. The Dutchman, Willem Einthoven started to do these measurements for the diagnosis of heartdiseases. He got the Nobel price for his innovative work. Einthoven described accurately how the measurements had to be performed by placing electrodes at three points on the body as shown in Figure 1: Right arm R, left arm L, and left leg F. The related electric potential differences were defined as:

- lead I: $L - R$
- lead II: $F - R$
- lead II: $F - L$

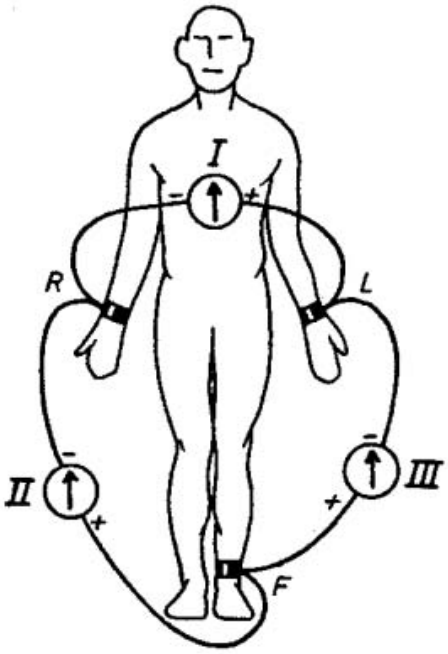


Figure 1: Einthoven method for measuring ECG.

In addition to the ECG-measurement you also perform a pulse-oxi measurement with the finger clip on your pointing finger. In the measuring program this is called *pleth*, the other signals associated with this diagnostic tool, like oxygen saturation and heart beat rate are not used.

Appendix B Equipment and software

The equipment set consists of the following hardware and software:

- ECG-amplifier and digitizer, coupled via a fibre to an USB-interface.
- Power supply for the ECG-amplifier
- Laptop with the TMSi PolyBench Data software (Icon on desktop).

- Finger clip with interface cable (attached to channel 16).
- Four single signal cables with a button coupling to be connected to the electrodes on your skin. (channel 1, 2, 3 and ground)
- Self-glueing electrodes are available in bags.

Appendix C Measurement procedure

C.1 Preparing for measurements

- Before you can start the measurements you have to place the ECG electrodes on a smooth area of skin on your wrists and ankles.
- Attach the finger clip for the pulse-oxi measurement to your pointing finger. Be sure to have the finger nail sign on top.

C.2 Acquisition of data

For acquiring the data, you have to start the TMSi Polybench Data software.

- A pop-up window will ask you for a password: `0000`.
- Select the patient in the patient window on the left (see Fig. 2); something like: `AT-TN ECG xx-xx-xxxx`
- Select the measurement in the right pane[1]; something like:
  - `AT-TN_Mod5_1024Hz_Poly5_32bit_v16` (for binary file)
  - `AT-TN_Mod5_1024Hz_CSV_32bit_v19` (for comma-separated-value file)

  a new window will appear with a graph and a few buttons (see Fig. 3).
- Press the button `start viewer`; this will start the measurement for orientation. You can optimize the graphical output via the *Auto Scale* and *Auto Offset* buttons.
- If you are ready to measure, press `Start Recording`.
- If needed: Use right-click on the signal names to (re-)set the scaling to auto-scaling.
- Press `Stop Recording` if you want to finish this measurement. Main window looks like in Fig. 4 after the measurement.

The data collected between `Start Recording` and `Stop Recording` will be saved automatically in the folder: `C:\work\AT TN M5\Measurements\ModATT` ≪`time-stamp`≫ \ with file names:

- ≪ filename ≫`.S00` – for a binary data file, see Fig. 5.
- ≪ filename ≫`.csv` – for a comma-separated data file, see Fig. 6.

For data handling you might need two Matlab files or a Python file to readout data from a binary file. However you can also chose a comma-separated-value format for general data-processing.

For loading and reading the binary files use the m-files `tms_plot.m` and `tms_read.m` to have a quick view of the data and check its usefulness. Inspection of the comma-separated file can be done directly via an editor or Excel.

[1]Both protocols contain the same measurement procedure and differ only in the format of data saving. For Matlab, scripts are available to read the binary data. For Python a concept script is available, however not fully tested yet. For Mathematica no binary read script is available. Both files contain the same data, however the binary file contains a header with some information about the measurements. Most important is the sampling rate of the measurement that is in both cases 1024 Hz

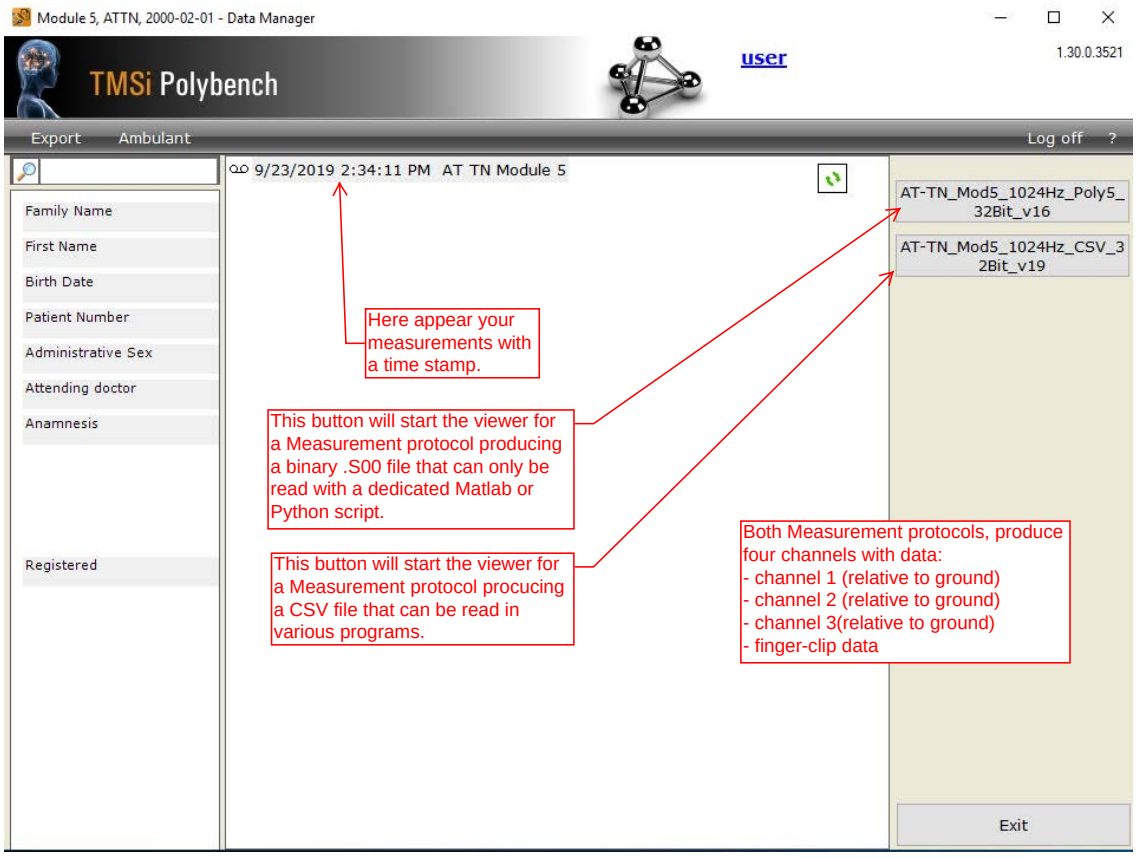


Figure 2: PolyBench Main window to select measurement protocol.

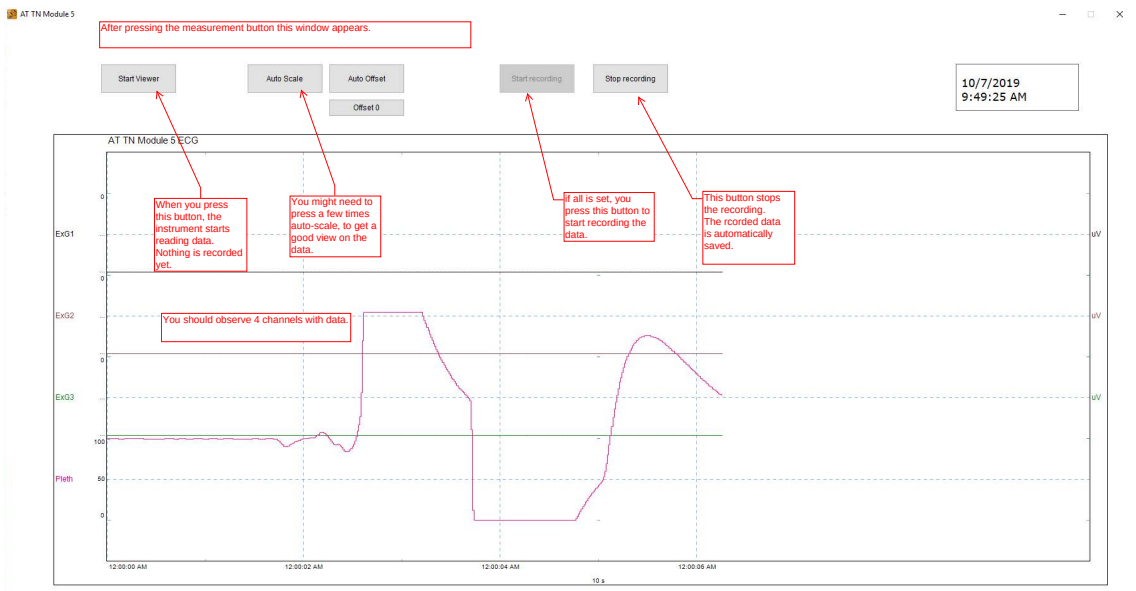


Figure 3: PolyBench Viewer to monitor the signals, adjust the settings and start/stop the recording. Data is automatically saved.

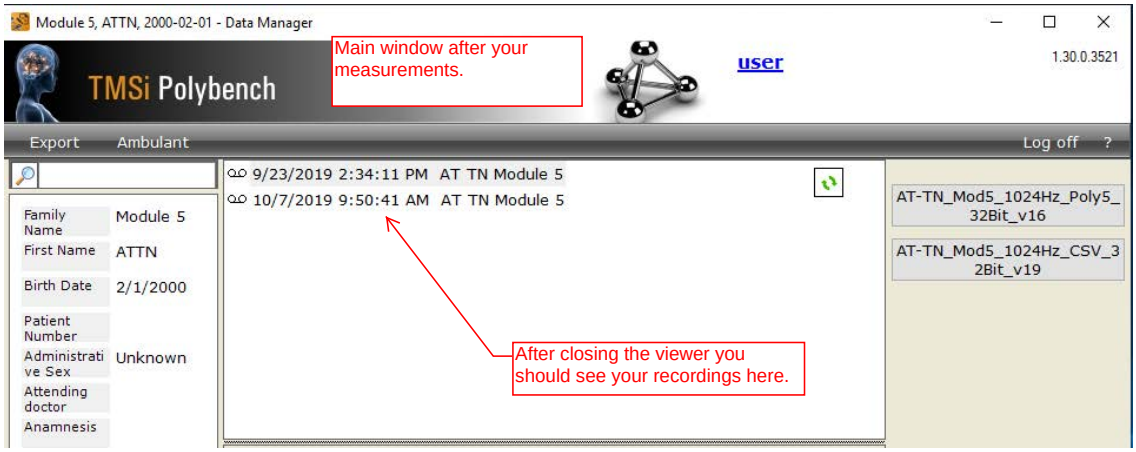


Figure 4: PolyBench Main window after measurement recording. The new recoding appears in the list.

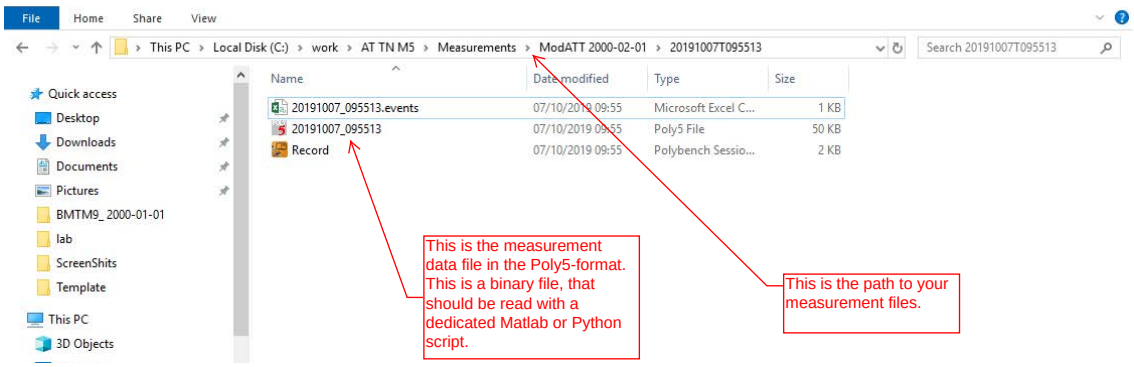


Figure 5: PolyBench data automatically saved: explorer view for Poly5-format (≪filename≫.S00).

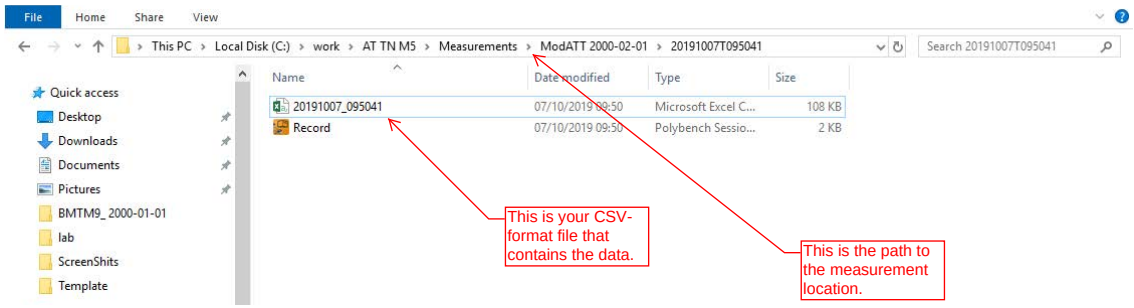


Figure 6: PolyBench data automatically saved: explorer view for CSV-format (≪filename≫.csv).